



Liver Function Tests

Biochemistry GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which aminotransferase is more specific for liver injury, and where else is the other one found?
2. What does AST > ALT suggest? What about ALT > AST?
3. Name four causes of a prolonged PT/INR.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

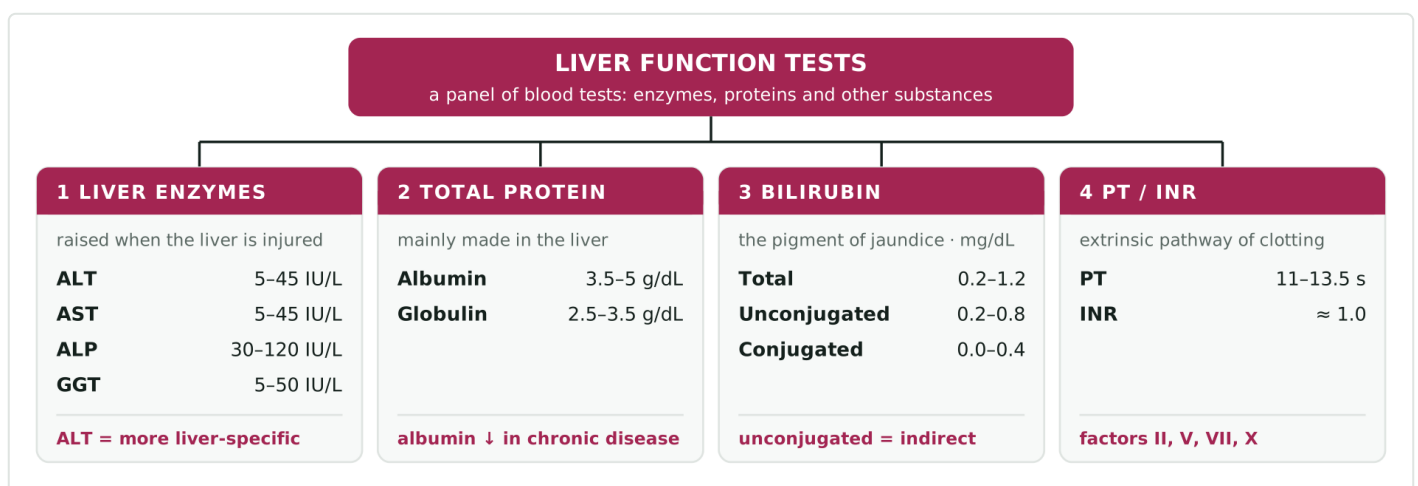
1 The liver and why we test it

The liver	Large organ in the upper right quadrant of the abdomen; weighs about 1.5 kg ; performs over 500 vital functions
LFTs	Also called liver chemistries — a panel of blood tests that measure enzymes, proteins and other substances to assess liver health and function
Used to	Diagnose liver diseases, monitor their severity, and check for liver damage caused by different conditions

Nine main functions of the liver

1 Digestion	Produces bile — essential for dietary fat digestion and absorption	6 Immune function	Filters bacteria and other pathogens from the bloodstream
2 Metabolism	Regulates metabolism of carbs, lipids and proteins	7 Hormone regulation	Activates and inactivates certain hormones, keeping hormonal balance
3 Detoxification	Converts harmful substances into harmless ones that can be excreted	8 Waste removal	Converts harmful substances into ones excreted through bile or urine
4 Storage	Glycogen ; vitamins A, D, E, K, B12; minerals (iron and copper)	9 Energy regulation	Stores, releases and produces glucose ; converts fats and proteins into energy when needed
5 Synthesis	Essential proteins and clotting factors ; carbohydrates, lipids, bile salts, hormones		

2 The panel at a glance



Concept map of the lecture's four test groups with their normal ranges — learn the map, then fill in the detail below.

3 Liver enzymes — the core discriminator

Enzyme	Reaction it catalyses	Found in	Normal	Raised in
ALT alanine aminotransferase	Transfers the amino group from alanine to α-ketoglutarate	Mainly the liver — more specific for liver injury than AST	5–45 IU/L	Very high levels: viral hepatitis, ischemic hepatitis , or injury from drugs or other chemicals
AST aspartate aminotransferase	Transfers the amino group from aspartate to α-ketoglutarate	Mostly the liver, but also heart, kidneys, skeletal muscle and brain — less liver-specific	5–45 IU/L	Liver damage (hepatitis, cirrhosis), MI , muscular dystrophy, kidney disease
ALP alkaline phosphatase	Hydrolyses phosphate groups from various compounds at an alkaline pH	Liver, bones, bile ducts and placenta — a non-specific marker of liver disease	30–120 IU/L	Liver inflammation, blockage of the bile ducts , bone diseases
GGT γ-glutamyl transferase	Transfers the γ-glutamyl group from glutathione to an amino acid	Mainly made in the liver, kidney and biliary tract; important in glutathione metabolism and recycling	5–50 IU/L	Liver and bile duct disorders, chronic alcohol consumption

AST : ALT NOTE

AST > ALT → alcoholic liver disease · **ALT > AST** → viral or non-alcoholic liver disease

4 Total protein — albumin and globulin

Total protein measures albumin + globulins; these proteins are mainly made in the liver.

Protein	What it is	Made by	Normal	Changes
Albumin	The most abundant protein synthesised by the liver	Liver	3.5–5 g/dL	Falls in all chronic liver diseases (cirrhosis, hepatitis, liver cancer); high levels are rarely due to liver problems
Globulin	A family of globular proteins — γ-globulins (immunoglobulins), α and β globulins, blood clotting globulins	Liver and immune system	2.5–3.5 g/dL	High γ-globulins : chronic liver diseases (hepatitis and cirrhosis), infection, multiple myeloma

5 Bilirubin

What it is	Byproduct of red blood cell breakdown ; the yellowish pigment seen in jaundice
Total (Tb)	0.2–1.2 mg/dL
Unconjugated	= free = indirect ; a bilirubin-albumin complex; 0.2–0.8 mg/dL
Conjugated	= direct ; 0.0–0.4 mg/dL
High in	Jaundice — mainly from gallstones, and acute or chronic hepatitis

6 Prothrombin time (PT) / INR

Evaluates	The extrinsic pathway of coagulation
Depends on	Clotting factors II, V, VII and X — the slide states all are made in the liver and require vitamin K
PT	Time for a blood sample to clot, in seconds — normal 11–13.5 s
INR	Standardised way of reporting PT, comparable across labs — normal ≈ 1.0
Prolonged in	Liver diseases (hepatitis, cirrhosis, acute liver failure) · vitamin K deficiency · clotting factor deficiency · warfarin therapy

7 Reading the panel — the lecture's three cases

Case	AST	ALT	ALP	Total bilirubin	Read-out
1 Healthy 21-year-old, routine test	22	25	90	0.7	All within the normal range → normal liver function
2 Yellow discoloration of the eyes	28	30	95	5 ↑	Bilirubin explains the jaundice; the enzymes are normal
3 45-year-old man, abdominal pain + yellow eyes	90 ↑	100 ↑	500 ↑	6 ↑	Cholestasis / obstructive jaundice (gallstone obstruction)

Enzymes in IU/L, bilirubin in mg/dL. In case 3 the ALP is about four times its upper limit, while AST and ALT are about double theirs.

P Must-remember pearls

- **ALT** is found mainly in the liver, so it is the more specific marker. **AST** is also in heart, kidneys, skeletal muscle and brain.
- **AST > ALT** → alcoholic liver disease. **ALT > AST** → viral or non-alcoholic liver disease.
- **ALP is non-specific** — liver, bones, bile ducts, placenta. GGT rises in liver and bile duct disorders and with chronic alcohol consumption.
- **Albumin falls** in all chronic liver diseases; **γ-globulins rise** in chronic liver disease, infection and multiple myeloma.
- **A prolonged PT/INR is not always the liver** — also vitamin K deficiency, clotting factor deficiency and warfarin therapy.

Q Test yourself

1. Yellow eyes. AST 28, ALT 30, ALP 95 IU/L; bilirubin 5 mg/dL. Which test explains the jaundice?

- A) ALT
B) Bilirubin
C) ALP
D) AST

2. Pain and yellow eyes. AST 90, ALT 100, ALP 500 IU/L; bilirubin 6 mg/dL. What is suspected?

- A) Normal liver function
B) Alcoholic liver disease
C) Vitamin K deficiency
D) Cholestasis / obstructive jaundice

3. After a viral illness, a woman has an ALT clearly higher than her AST. What does this suggest?

- A) Viral or non-alcoholic liver disease
B) Alcoholic liver disease
C) Bone disease
D) Multiple myeloma

4. A man has had cirrhosis for ten years. Which result is expected?

- A) High albumin
B) Low γ-globulins
C) Low albumin
D) A shortened PT

5. A man with bone disease has ALP 380 IU/L and normal bilirubin. Why can ALP rise here?

- A) It is found mainly in the heart
B) It is non-specific and also found in bone
C) It is a clotting factor
D) It is the most liver-specific enzyme

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1. **B — Bilirubin** — only bilirubin is above its range (0.2–1.2 mg/dL).
2. **D — Cholestasis / obstructive jaundice** — ALP and bilirubin are high; the lecture gives gallstone obstruction.
3. **A — Viral or non-alcoholic liver disease** — ALT > AST; the reverse (AST > ALT) points to alcohol.
4. **C — Low albumin** — albumin falls in all chronic liver diseases; γ-globulins rise and PT lengthens.
5. **B — It is non-specific and also found in bone** — ALP is in liver, bones, bile ducts and placenta.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam